Genuine Multipartite Entanglement Analysis

Hubbard ring - Convex-roof framework

System parameters

Model	**Hubbard 1D-PBC**
Mode	**uniform**
Sites (L)	**6**
Electrons (Ne)	**6**
Hopping (t)	**1.0**
Interaction (U)	**4.0**
n-body order	**5**
Eigenstate idx	**0 (ground state)**

Eigenstate properties

Energy E0	**-3.66870618**
Energy E1	**-2.89838147**
Gap (E1-E0)	**0.770325**
<S^2>	**-0.000000**
S	**-0.0000**
<Sz>	**0.000000**
Multiplicity	**1.0**
Spin label	**Singlet (S=0)**

Double occupancy

<D> total	**0.666395**
<D> / site	**0.111066**

RDM and GME statistics for this run (5-body clusters)

Hilbert space dim : 2^5 = 32
Number of clusters : 6
Pure clusters : 0
Mixed clusters : 6
Clusters with GME (>0) : 6

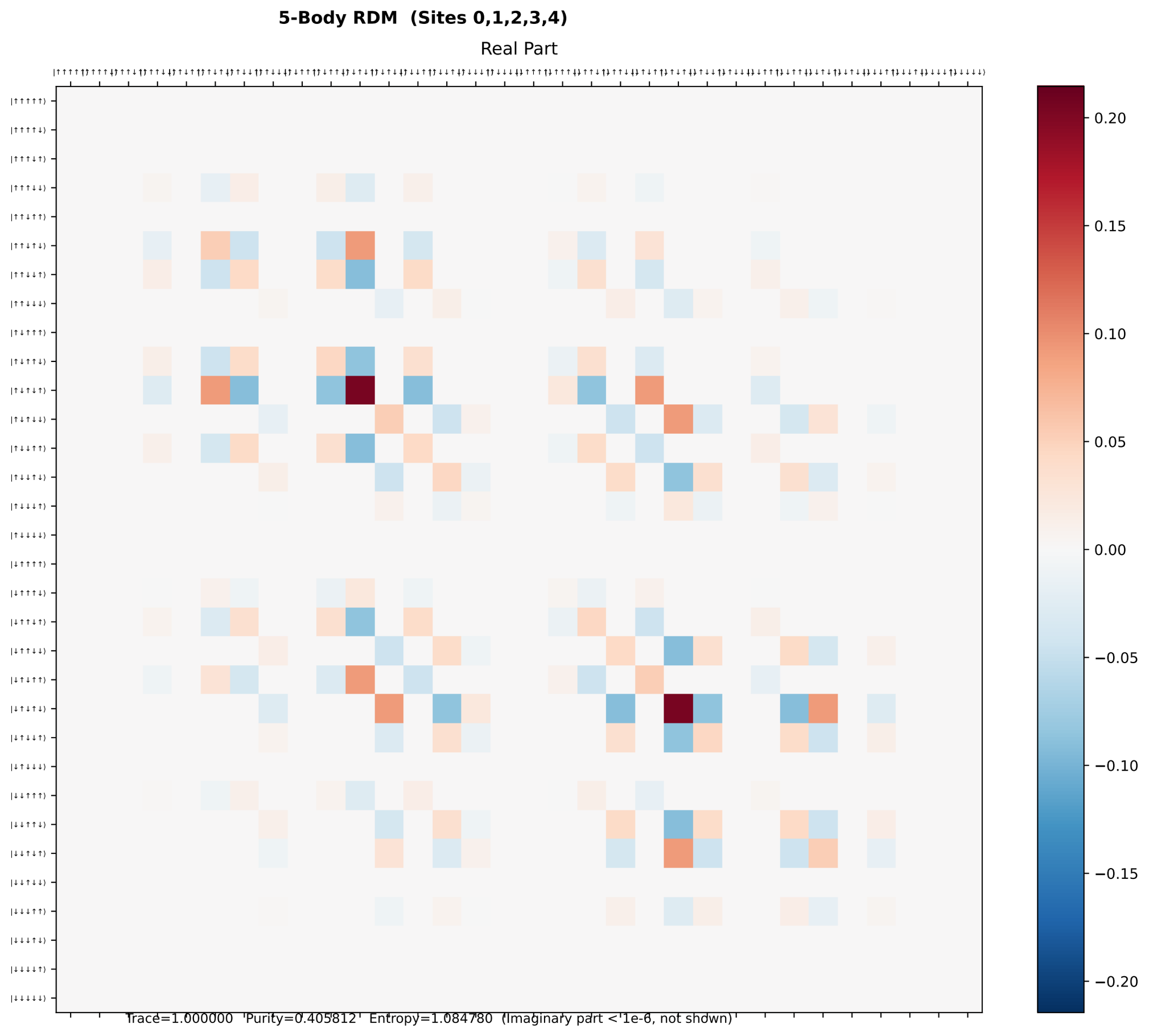
Purity gamma = Tr(rho^2) : mean = 0.405812 [min 0.405812 , max 0.405812]
Von Neumann entropy : mean = 1.084780 [min 1.084780 , max 1.084780]
Rank of rho : mean = 10.00 [min 10 , max 10]
C_GME (best estimate) : mean = 0.403898 [min 0.401753 , max 0.405361]

Summary: All 5-Body RDM Clusters

System: L=6, Ne=6, U=4.0, t=1.0 | E_0=-3.668706

Sites	Trace	Purity	Entropy	C_GME	C_LB	C_gcg	C_stiefel_gcg	C_stiefel	C_UB	Source	Status
(0, 1, 2, 3, 4)	1.000000	0.405812	1.084780	–	0.000000	0.397937	0.397994	0.404057	0.757393	S:NEW G:NEW SG:NEW	MIXED
(1, 2, 3, 4, 5)	1.000000	0.405812	1.084780	–	0.000000	0.397993	0.401487	0.405361	0.757393	S:NEW G:NEW SG:NEW	MIXED
(2, 3, 4, 5, 0)	1.000000	0.405812	1.084780	–	0.000000	0.401452	0.401478	0.405275	0.757393	S:NEW G:NEW SG:NEW	MIXED
(3, 4, 5, 0, 1)	1.000000	0.405812	1.084780	–	0.000000	0.401487	0.401485	0.401753	0.757393	S:NEW G:NEW SG:NEW	MIXED
(4, 5, 0, 1, 2)	1.000000	0.405812	1.084780	–	0.000000	0.401487	0.401471	0.404291	0.757393	S:NEW G:NEW SG:NEW	MIXED
(5, 0, 1, 2, 3)	1.000000	0.405812	1.084780	–	0.000000	0.397990	0.397930	0.402653	0.757393	S:NEW G:NEW SG:NEW	MIXED

Legend:
  Trace = Tr(rho) | Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_gcg = unitary optimization - exponential-map GCG | BEST = selected estimate
  Source: S=Stiefel, G=GCG, SG=Stiefel-GCG; PKL=reused from pkl, NEW=recomputed, MISS->NEW=pkl missing so recomputed
  C_UB = Upper bound (eigendecomp) | Status: GME=has GME | NO GME=separable | MIXED=mixed state


5-Body RDM (Sites 0,1,2,3,4)
Real Part
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Trace=1.000000 Purity=0.405812 Entropy=1.084780 (Imaginary part <1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 0,1,2,3,4)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.4058117098
Von Neumann entropy	1.0847796368
Eigenvalue min	-2.638645e-17
Eigenvalue max	4.486714e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.40581171

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

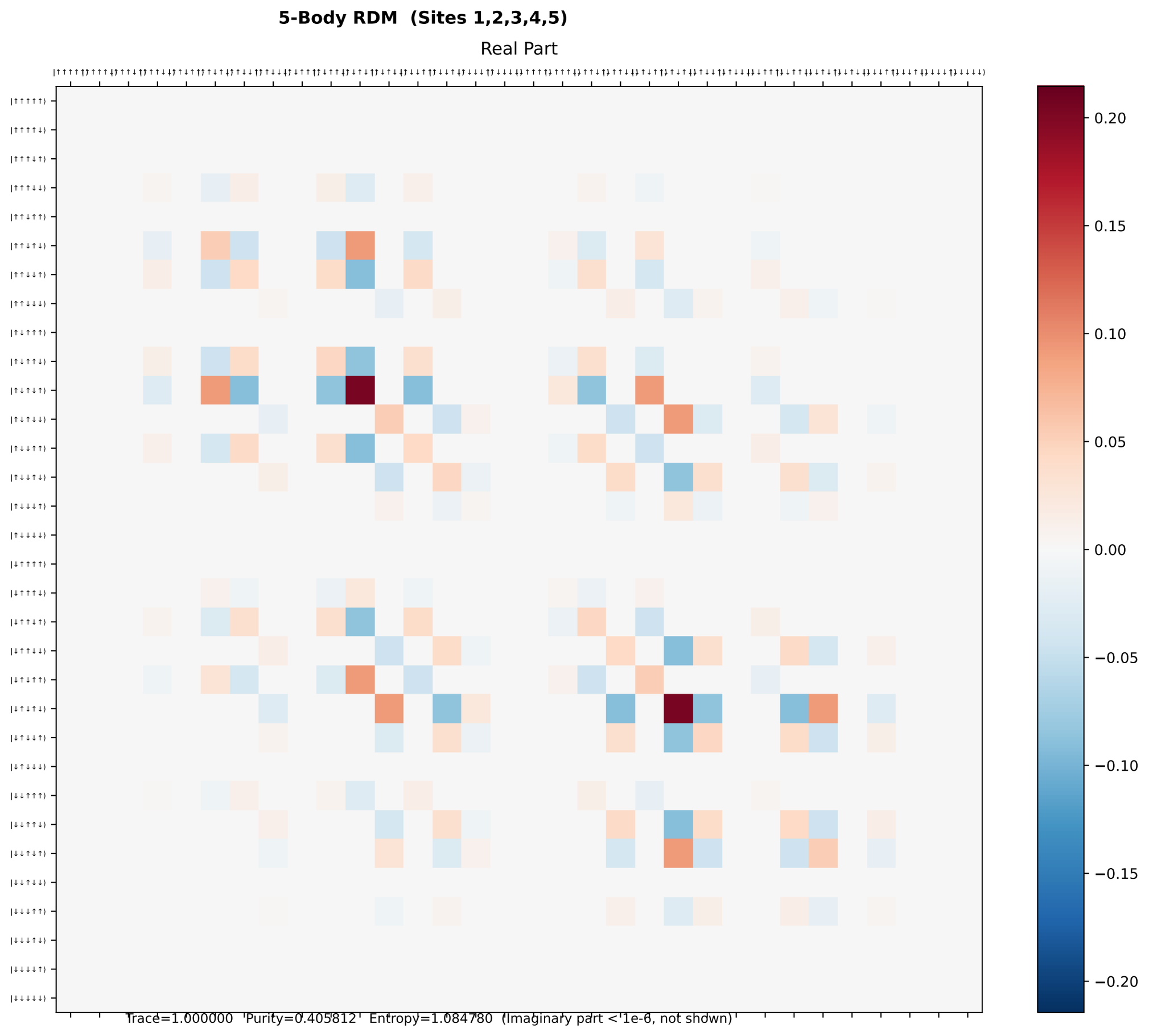
Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

Convex Roof Analysis
==========

Lower Bounds:
C_LB = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = 0.40405694 BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.75739264

Status: nONE

Solver source:
stiefel: NEW gcg: NEW stiefel_gcg: NEW


5-Body RDM (Sites 1,2,3,4,5)
Real Part
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Trace=1.000000 Purity=0.405812 Entropy=1.084780 (Imaginary part <1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 1,2,3,4,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4058117098
  Von Neumann entropy  : 1.0847796368
  Eigenvalue min       : -6.481893e-17
  Eigenvalue max       : 4.486714e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.40581171

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------------------

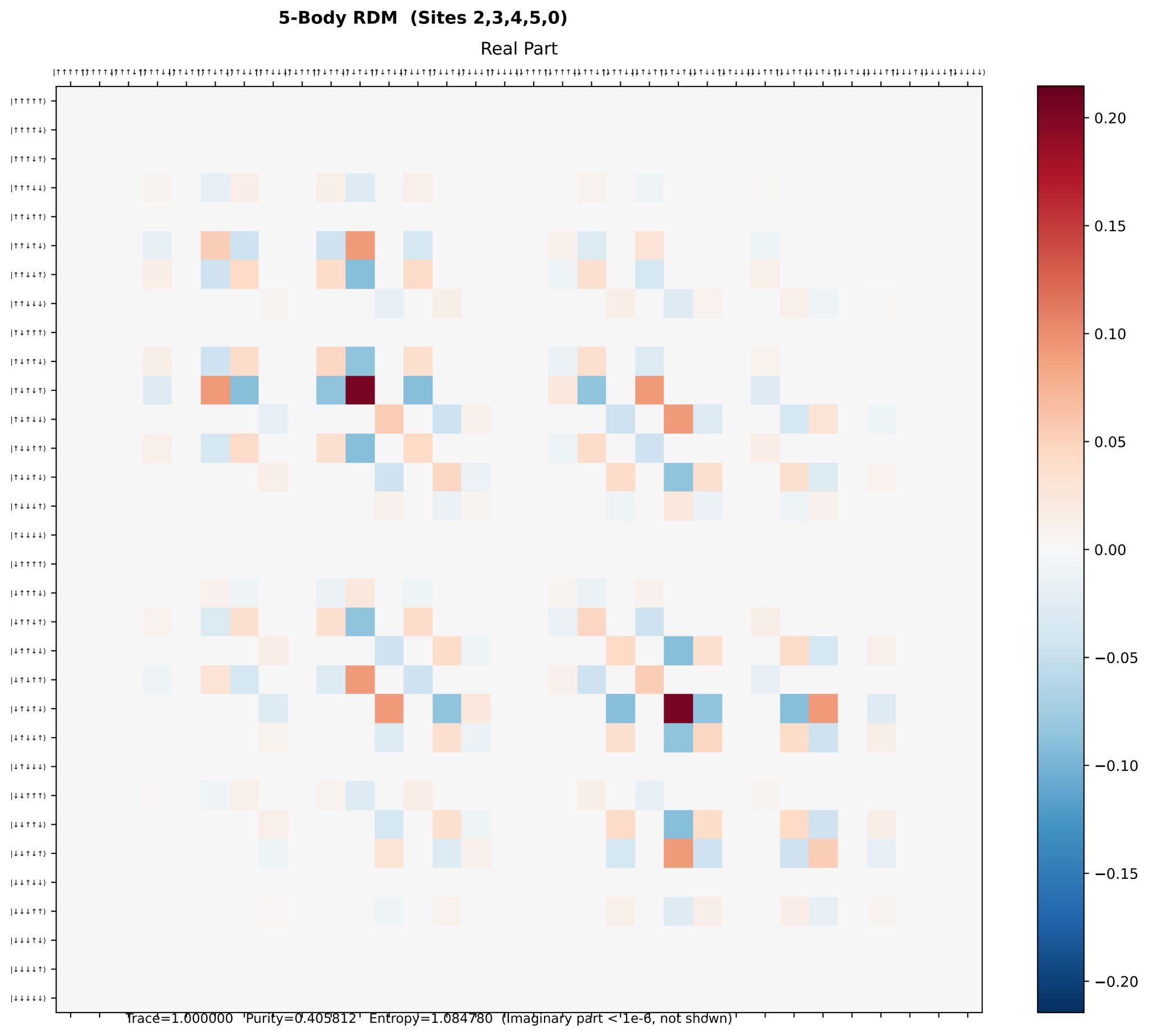
Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.40536133   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.75739264

  Status: nONE

  Solver source:
    stiefel: NEW  gcg: NEW  stiefel_gcg: NEW
```


5-Body RDM (Sites 2,3,4,5,0)
Real Part
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Trace=1.000000 Purity=0.405812 Entropy=1.084780 (Imaginary part <1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 2,3,4,5,0)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4058117098
  Von Neumann entropy  : 1.0847796368
  Eigenvalue min       : -2.848592e-17
  Eigenvalue max       : 4.486714e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.40581171

  A            | B            | Tr(rho_A^2)
  ____________ | ____________ | ___________

STATUS: MIXED STATE
_____________________________

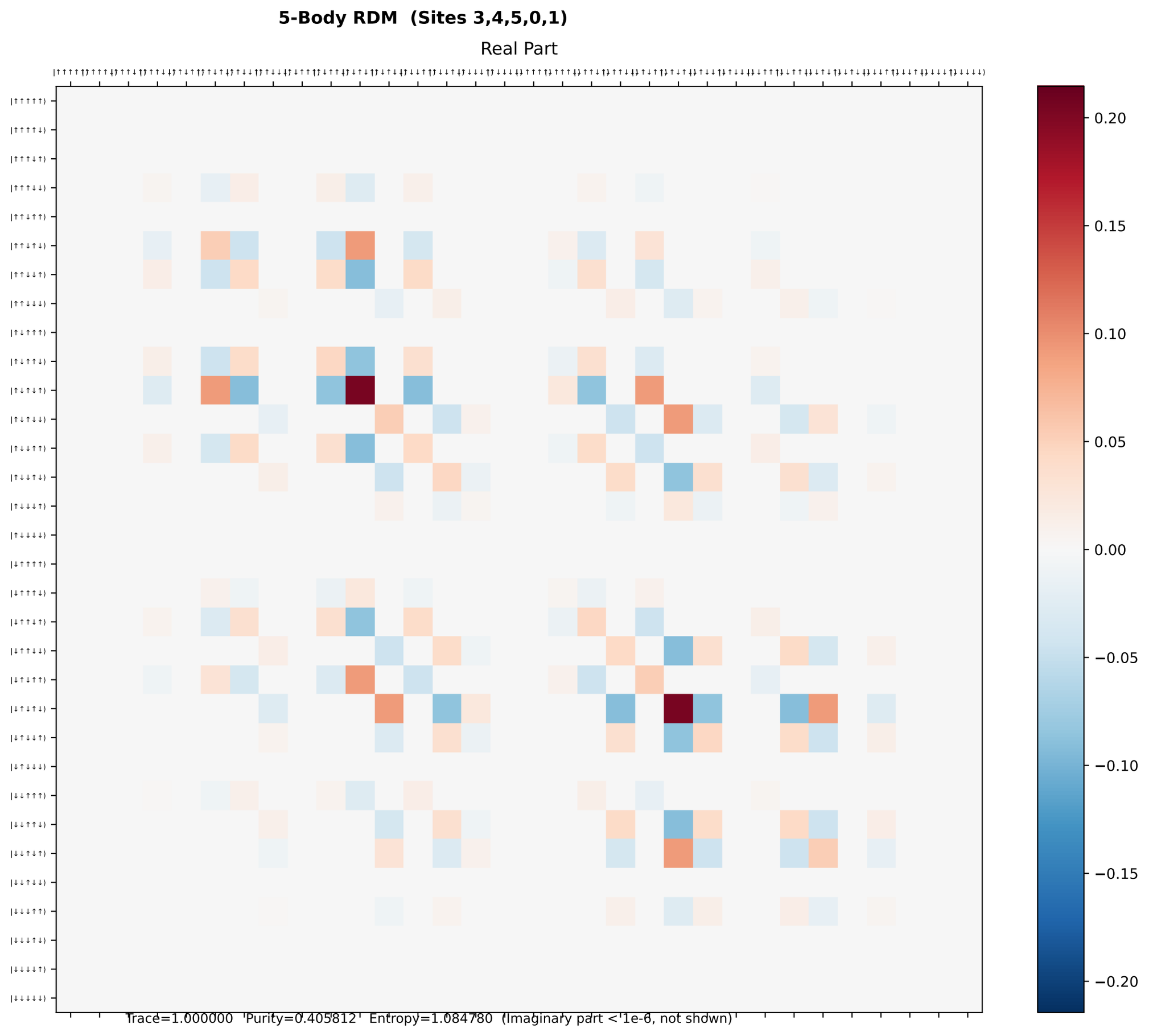
Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.40527506   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.75739264

  Status: nONE

  Solver source:
    stiefel: NEW  gcg: NEW  stiefel_gcg: NEW
```


5-Body RDM (Sites 3,4,5,0,1)
Real Part
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Trace=1.000000 Purity=0.405812 Entropy=1.084780 (Imaginary part <1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 3,4,5,0,1)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4058117098
  Von Neumann entropy  : 1.0847796368
  Eigenvalue min       : -8.529563e-17
  Eigenvalue max       : 4.486714e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.40581171

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
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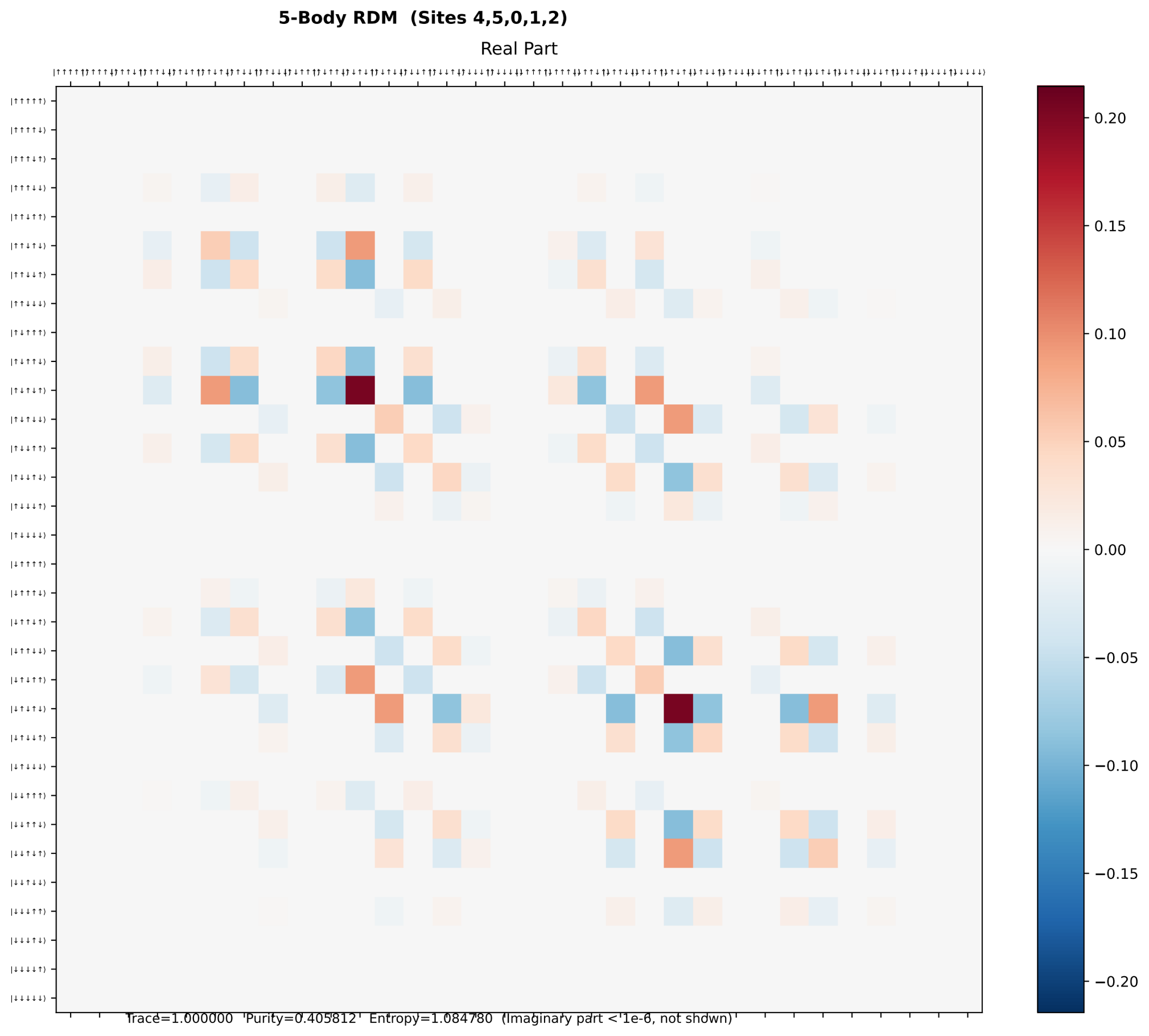
Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.40175314   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.75739264

  Status: nONE

  Solver source:
    stiefel: NEW  gcg: NEW  stiefel_gcg: NEW
```


5-Body RDM (Sites 4,5,0,1,2)
Real Part
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Trace=1.000000 Purity=0.405812 Entropy=1.084780 (Imaginary part <1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 4,5,0,1,2)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4058117098
  Von Neumann entropy  : 1.0847796368
  Eigenvalue min       : -4.241866e-17
  Eigenvalue max       : 4.486714e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.40581171

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
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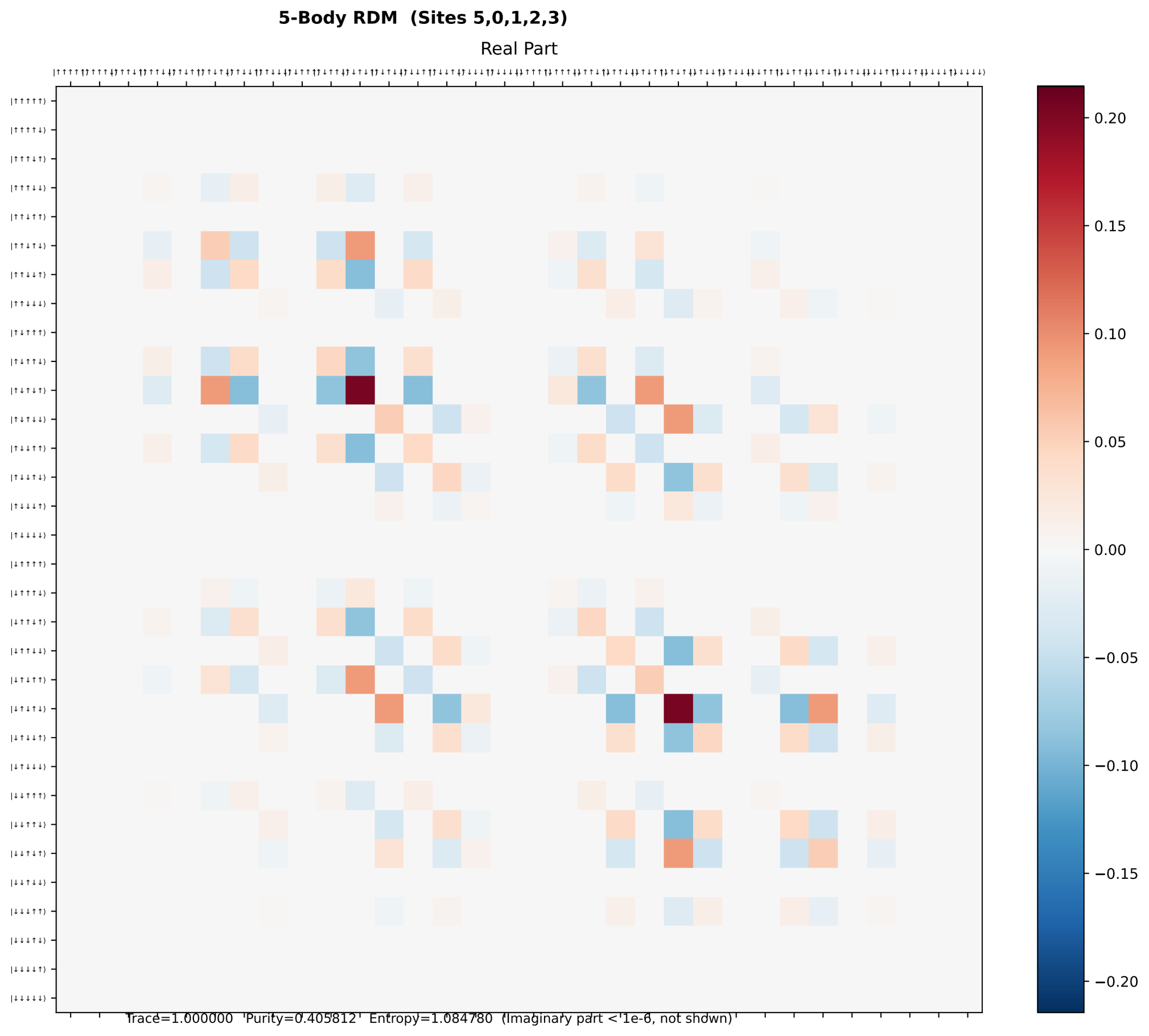
Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.40429123   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.75739264

  Status: nONE

  Solver source:
    stiefel: NEW  gcg: NEW  stiefel_gcg: NEW
```


5-Body RDM (Sites 5,0,1,2,3)
Real Part
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Trace=1.000000 Purity=0.405812 Entropy=1.084780 (Imaginary part <1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 5,0,1,2,3)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4058117098
  Von Neumann entropy  : 1.0847796368
  Eigenvalue min       : -2.962425e-17
  Eigenvalue max       : 4.486714e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.40581171

  A            | B            | Tr(rho_A^2)
  ____________ | ____________ | ___________

STATUS: MIXED STATE
____________________________________

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.40265294   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.75739264

  Status: nONE

  Solver source:
    stiefel: NEW  gcg: NEW  stiefel_gcg: NEW
```